

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 104		Department of Computing	

2nd SEMESTER 2024/25 FINAL EXAMINATION**OPERATING SYSTEMS CONCEPTS****TIME ALLOWED: 2 Hours****INSTRUCTIONS TO CANDIDATES**

1. This is an open-book examination.
2. Total marks available are 100, accounting for 80% of the overall module marks.
3. Answer all FOUR questions.
4. The number in the column on the right indicates the marks for each question.
5. Relevant and clear steps should be included in the answers.
6. The university approved calculator - Casio FS82ES/83ES can be used.
7. Only English solutions are accepted.
8. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

QUESTION I. Fundamentals

(35 marks)

1. How can the system distinguish between the pages that are in main memory from the pages that are on the disk? Explain your answer.

(6 marks)

2. Define the **semaphore** operations.

(6 marks)

3. What is a **virtual machine**? Why is this concept especially useful for those who manage large-scale computing systems?

(6 marks)

4. Which of the followings spend less time in **context switching**? Option A or B? Explain your answer.

A. Switching between two user-level threads in the same process.

B. Switching between two user-level threads in different processes.

(6 marks)

5. Explain what an **Access Matrix** in computer protection is.

(6 marks)

6. Why do all File Systems suffer from **internal fragmentation** (disk)?

(5 marks)

QUESTION II. CPU scheduling, Memory management, and Disk scheduling**(41 marks)**

1. Consider the following scenario of processes. Their arrival time and burst time are as follows:

Process	Arrival time (ms)	Burst time (ms)
P1	0	8
P2	2	6
P3	4	1
P4	1	9
P5	3	3

Draw the **Gantt chart** for the execution of the processes and calculate the **turnaround time** for each process using the **Shortest-Job-First (SJF)** algorithm.

(7 marks)

Draw the **Gantt chart** for the execution of the processes and calculate the **turnaround time** for each process using the **Shortest-Remaining-Time-First (SRTF)** algorithm.

(7 marks)

2. Calculate the number of page faults for the following sequence of page references (each element in the sequence represents a page number) using the **Least Recently Used (LRU)** replacement algorithm with frame size of 3.

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

(6 marks)

3. Consider a disk queue with I/O requests on the following cylinders in their arriving order:

6, 10, 12, 54, 97, 73, 128, 15, 44, 110, 34, 45

The disk head is assumed to be at cylinder 23 and moving in the direction of decreasing number of cylinders. The disk consists of 150 cylinders.

Write the sequence in which requested tracks are serviced using **LOOK algorithm** and calculate the **total head movement** (in number of cylinders) incurred while servicing these requests.

(6 marks)



4. A paging scheme uses a **Translation Lookaside buffer (TLB)**. A TLB access takes 30 ns and a main memory access takes 70 ns. What is the **Effective Access Time** (in ns) if the TLB hit ratio is 75% and there is no page fault?

(6 marks)

5. You have a computer that has the following specifications:

- i. 12 bits virtual addresses (VA)
- ii. Each page has 16 bytes (paging system)
- iii. The Main Memory (RAM) has 256 bytes.

Answer the following questions:

- (a) What is the size of the virtual address space? (3 marks)
- (b) How many bits in the VA are used as the Virtual Page Number (VPN)? (3 marks)
- (c) How many page frames does the RAM have? (3 marks)

QUESTION III. Resource allocation

(12 marks)

Consider the following snapshot of a system:

Available

A	B	C	D
1	0	1	0

Process	Max				Allocation			
	A	B	C	D	A	B	C	D
P0	0	1	2	3	0	1	1	2
P1	2	0	0	2	1	0	0	2
P2	0	2	0	1	0	1	0	1
P3	2	2	0	0	1	1	0	0
P4	0	2	2	0	0	1	2	0

a) What is the content of the **Need** matrix denoting the number of resources needed by each process? (2 marks)

b) Is this system in a **safe state**? If your answer is *yes*, please give a safe sequence and resources available after each process finished. If your answer is *no*, please specify the processes that might involve in a deadlock (unsafe). (5 marks)

c) If a request from **P3** arrives for **(1, 0, 0, 0)**, can that request be safely granted immediately? Explain your answer. (5 marks)

**QUESTION IV. Operating System in C Language****(12 marks)**

The following code implements a ticket booking system where multiple threads attempt to book seats from a shared pool of 100 seats.

Three semaphores are used: *mutex* (initialized to 1), *seats_available* (initialized to 100), and *booking_done* (initialized to 0).

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define TOTAL_SEATS 100
sem_t mutex, seats_available, booking_done;
int seats = TOTAL_SEATS;

void *book_seat(void *arg) {
    int id = *(int *)arg;
    sem_wait(&booking_done);
    sem_wait(&mutex);
    seats--;
    printf("Thread %d booked a seat. Remaining seats: %d\n", id, seats);
    sem_post(&mutex);
    sem_post(&seats_available);
    pthread_exit(NULL);
}

void *release_seat(void *arg) {
    int id = *(int *)arg;
    sem_wait(&seats_available);
    sem_wait(&mutex);
    seats++;
    printf("Thread %d released a seat. Remaining seats: %d\n", id, seats);
    sem_post(&mutex);
    sem_post(&booking_done);
    pthread_exit(NULL);
}
```

a) Explain the purpose of *seats_available* and *booking_done* semaphores.

(4 marks)

b) The current implementation ensures that the *seats* variable is protected by the *mutex* semaphore. However, there is still a potential issue with the order of semaphore operations. Identify the issue and propose a fix. (4 marks)

c) Suppose `sem_post(&booking_done)` is replaced with `sem_post(&seats_available)` in `release_seat()`. What issue would arise? (4 marks)

END OF EXAM PAPER